Note on Over-simplified Cross-Situational Learning Models

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April 22, 2016

Abstract

1 Blythe, Smith, & Smith (2010)

1.1 On the fast-mapping learner

Suppose that the fast-mapping learner learns a word independently drawn from uniform distribution of W words given in each episode. As every episode has one word by the probability $1/W$, this is identified with so called Coupon Collector's Problem. In this problem, the expectation of the time to sample all the words T is

$$E[T] = \sum_{i=1}^{W} E[t_i].$$

where t_i is the time to sample a i^{th} new word given $(i-1)$ words being learned. Thus,

$$T = W \sum_{i=1}^{W} 1/i \approx W \log W.$$

1.2 Extension of the fast-mapping model

This estimate by Blythe et al. is a quite "optimistic", as the target word is sampled uniformly. Now let us consider a more realistic case – the word is drawn from a Zipf's distribution (or power distribution). Suppose that the i^{th} word is drawn by the probability $p_i \geq 0$, and the there are n words and $\sum_{i=1}^{n} p_1 = 1$. Write $p = (p_1, \ldots, p_n)$. According to Hidaka (2014), the probability to have the number of unique words (types) k for m sampled words (tokens) is, as $m \to \infty$,

$$P(k \mid m, p) \approx Q_k(p_m),$$

where $Q_k(p_m)$ is the Poisson-binimial distribution with the parameter $p_m := (p_{1,m}, p_{2,m}, \ldots, p_{n,m})$ and $p_{i,m} := 1 - (1 - p_i)^m$.

Suppose, for a small $\epsilon > 0$, the probability to complete sample n types satisfies

$$\prod_{i=1}^{n} p_{i,m} > 1 - \epsilon.$$

Using this inequality with $\epsilon = 0.01$, which Blythe et al assumed in their estimate, we have the results for example:

1. (Uniform sampling) For the set of $n = 10,000$ words drawn by the uniform distribution $p_i = 1/n$ and and $\epsilon = 0.01$, the minimal m satisfies the inequality above is ~~46,050~~138,098.

2. (Zipf's law) For the set of $n = 10,000$ words drawn by the Zipf's distribution $p_i \propto 1/i$ and $\epsilon = 0.01$, the minimal m satisfies the inequality above is 1,099,950.

The first case is considered in Blythe et al. (2010), and they argued that this is a reasonable estimate with respect to empirical findings in developmental literature. The second case, in which I additionally relax their uniform sampling assumption, offers a "less reasonable" estimate, which is over 20 times more number of samples to finish learning than the uniformly distributed case. This estimate may break down their theory: As the fast-mapping estimate is the minimal baseline required to learn words, their assumption that independent learning is shown quite unreasonable to begin with.

2 Appendix

For a set of N words drawn from the distribution $p = (p_1, \ldots, p_N)$, the proportion of children finished learning all the words is $(1 - \epsilon)$ for $0 < \epsilon < 1$

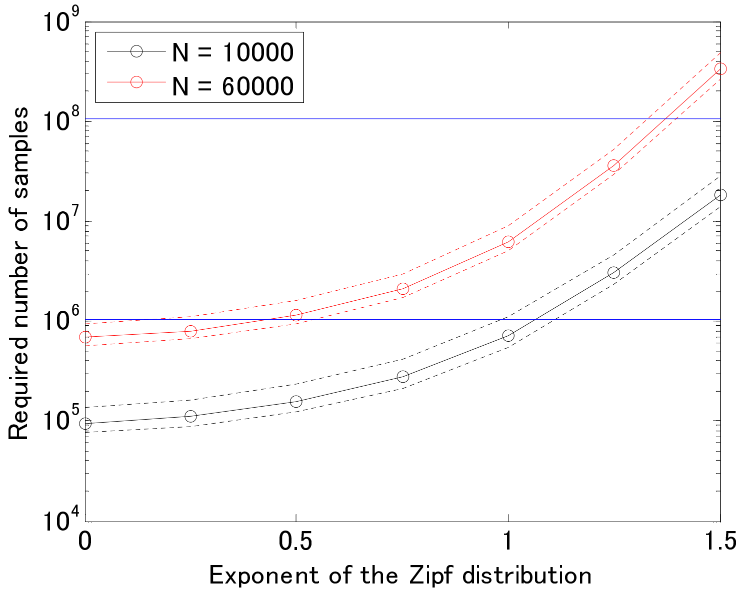

Figure 1: For $\epsilon = 0.01, 0.5, 0.99$ (broken and solid lines), $N = 10000, 60000$ and the exponent $a = 0, 0.25, \ldots, 1.5$, the required number of samples M for a generalized Zipf distribution $p_k = k^{-a}/\sum_{k=1}^{N} k^{-a}$ is numerically calculated by the root of Equation (1).

requires the sample size M, which is the root of

$$\prod_{i=1}^{N} \left(1 - (1 - p_i)^M\right) = 1 - \epsilon. \tag{1}$$

Write

$$f_{N,\epsilon}(x) := \frac{\log\left(1 - (1-\epsilon)^{1/N}\right)}{\log(1-x)}.$$

In a special case with a set of N words distributed uniformly, the proportion of children finished learning all the words is $(1 - \epsilon)$ requires the sample size

$$M = f_{N,\epsilon}(1/N).$$

By the intermediate value theorem, the root of (1) is expressed with some constant $\min p \leq c \leq \max p$

$$M = f_{N,\epsilon}(c).$$

Equivalently, we have inequality

$$f_{N,\epsilon}(\max p) \leq M \leq f_{N,\epsilon}(\min p).$$

Write $M_+ = f_{N,\epsilon}(\min p)$, which gives a reasonable estimate of the upper bound of the root M of (1). For the Zipf distribution with N words and exponent a, $\min p = N^{-a} H_{N,a}^{-1}$ where $H_{N,a} := \sum_{k=1}^{N} k^{-a}$ is the generalized harmonic number.

$$\frac{\partial f_{N,\epsilon}(x)}{\partial x} = -\frac{\log\left(1-(1-\epsilon)^{1/N}\right)}{(1-x)\log(1-x)^2} = -\frac{f_{N,\epsilon}(x)}{(1-x)\log(1-x)}.$$

$$\min p = \frac{N^{-a}}{\sum_{k=1}^{N} k^{-a}}$$

$$\frac{\partial}{\partial a}\min p = \frac{-N^{-a}\log N\left(\sum_{k=1}^{N} k^{-a}\right) - N^{-a}\left(-\sum_{k=1}^{N} k^{-a}\log k\right)}{\left(\sum_{k=1}^{N} k^{-a}\right)^2}$$

$$\frac{\partial}{\partial a}\min p = \min p \frac{\left(\sum_{k=1}^{N} k^{-a}\log\frac{k}{N}\right)}{\sum_{k=1}^{N} k^{-a}} = \min p\left(-\frac{\partial H_{N,a}}{\partial a}/H_{N,a} - \log N\right)$$

$$\frac{\partial \log M_+}{\partial a} = \frac{p_{\min}\left(\frac{\partial H_{N,a}}{\partial a}/H_{N,a} + \log N\right)}{(1-p_{\min})\log(1-p_{\min})}$$

and ....

$$\frac{\partial^2 \log M_+}{\partial a^2} \geq 0.$$

This implies the M_+ is a super-exponential monotone function.

2.1 Cross-situational learning

The idea used in the analysis of fast learning can be generalized to analyze the cross situation learning.

Suppose, for a particular target word in a set of N words, i^{th} other word appears by the probability q_i in each learning episode. Write $q =$

$(q_1, \ldots, q_{N-1})$. The probability to none of the $N-1$ other words acompanies with the target word for every time of k learning episodes is

$$P_{\mathrm{XSL}}(K, q) = \prod_{i=1}^{N-1} \left(1 - q_i^K\right).$$

Find this equation is essentially identical to the left hand side of (1) by identifying $K = M$ and $q_i = 1 - p_i$. Thus, the upper bound of the K is given by

$$K_+ = f_{N-1,\epsilon}(1 - \max q).$$

Putting together, the time for cross situational learning is bounded by

$$T_+ = \max_{p,q} f_{N,\epsilon}(p) f_{N-1,\epsilon}(1-q).$$

This analysis implies the following.

1. The uniform sampling of (1) target word and (2) incidental words gives shortest learning time.

2. The sampling of (1) rarely drawn target word acompanying with (2) highly likely incidental words gives longest learning time.

It is obvious that the uniform sampling gives the largest $\min p$ and smallest $\max q$. On the other hand, a class of Zipf distributions give relatively severe conditions for both fast mapping and cross situational learning, as it gives extreme values – smaller $\min p$ and larger $\max q$. Specifically, under the Zipf distribution with exponent a, the learning time for the worst case 2. is bounded by the super-exponential function of the a.

In summary, this analysis revealed that efficient cross situational learning needs to have every target word and its incidental distribution is near uniform distributions in both p and q. This condition required for the efficient word learning is hard to believe in with respect to empirical data (Figure 2)!

Acknowledgment

Thanks to George and Takuma, I could write this note.

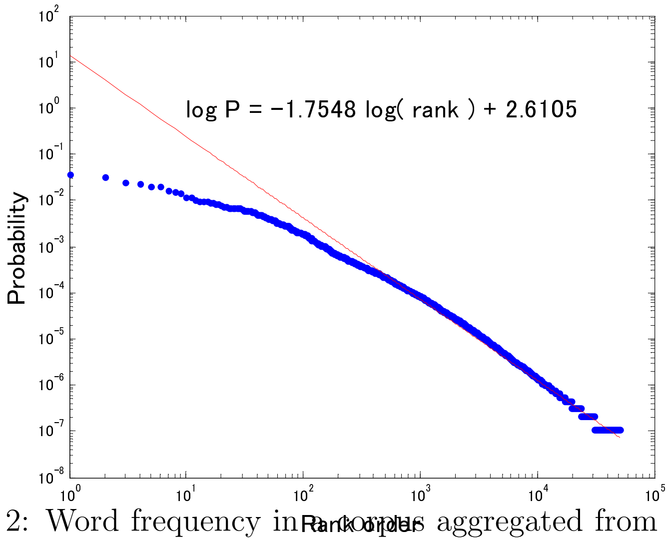

Figure 2: Word frequency in a corpus aggregated from the CHILDES transcripts.